Endoscopist de-skilling risk after exposure to artificial intelligence in colonoscopy: a multicenter observational study

Running title: AI-exposure in colonoscopy

Authors

Krzysztof Budzyń, MD [1,2], Marcin Romańczyk, MD [1,2], Diana Kitala, PhD [3], Paweł Kołodziej, MD [4], Marek Bugajski, MD [5], Hans Olov Adami, MD [6,7], Johannes Blom, MD [8], Marek Buszkiewicz, MD[9], Natalie Halvorsen, MD [6], Cesare Hassan, MD* [10,11], Tomasz Romańczyk, MD [1,2], Øyvind Holme, MD* [12,13], Krzysztof Jarus, MD [14], Shona Fielding, PhD [15], Melina Kunar, PhD [16], Maria Pellise, MD* [17, 18], Nastazja Pilonis, MD [6,19,20], Michał Filip Kamiński, MD* [6,21,22], Mette Kalager MD* [6], Michael Bretthauer, MD* [6], Yuichi Mori, MD* [6,23]

* Full professors

1. Department of Gastroenterology, Faculty of Medicine, Academy of Silesia, Katowice, Poland.
2. Endoterapia, H-T. Centrum Medyczne, Tychy, Poland.
3. Agency of Medical Research, Warsaw, Poland.
4. Endomed, Radom, Poland.
5. Polish Foundation of Gastroenterology, Warsaw, Poland.
6. Clinical Effectiveness Research Group, University of Oslo and Oslo University Hospital, Oslo, Norway.
7. Department of Medical Epidemiology and Biostatistics, Karolinska Institutet, Stockholm, Sweden.

8. Department of Surgery, Södersjukhuset, Stockholm and Department of Clinical Science and Education, Karolinska Institutet, Stockholm, Sweden.
9. Buszkiewicz Medical Center, Gorzow Wielkopolski, Poland.
10. Department of Biomedical Sciences, Humanitas University, Pieve Emanuele, Milan, Italy.
11. Endoscopy Unit, IRCCS Humanitas Research Hospital, Rozzano, Milan, Italy.
12. Institute of Health and Society, University of Oslo, Oslo, Norway.
13. Research Unit, Sorlandet Hospital, Kristiansand, Norway.
14. Centre of Gastroenterology, Wodzislaw Slaski. Poland.
15. Frontier Science (Scotland) Ltd, Kincraig, United Kingdom.
16. Department of Psychology, University of Warwick, Coventry, United Kingdom.
17. Department of Gastroenterology, Hospital Clínic Barcelona, Centro de Investigación Biomédica en Red en Enfermedades Hepáticas y Digestivas (CIBEREHD), IDIBAPS (Institut d'Investigacions Biomèdiques August Pi i Sunyer), Barcelona, Spain.
18. Facultat de Medicina i Ciències de la Salud, Universitat de Barcelona (UB), Barcelona, Spain.
19. Department of Oncological Gastroenterology, National Research Institute of Oncology, Warsaw, Poland.
20. Division of General, Endocrine and Transplant Surgery, Medical University of Gdańsk, Gdańsk, Poland.
21. Department of Gastroenterology, Hepatology and Clinical Oncology, Centre of Postgraduate Medical Education, Warsaw, Poland.
22. Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland.
23. Digestive Disease Center, Showa University Northern Yokohama Hospital, Yokohama, Japan.

Corresponding author

Marcin Romańczyk, MD, PhD

1. Department of Gastroenterology, Faculty of Medicine, Academy of Silesia, Katowice,

Poland.

2. Endoterapia, H-T. Centrum Medyczne, Tychy, Poland.

Aleja Bielska 105, 43-100 Tychy, Poland

Tel: +48 506516465

Email: marcin.romanczyk@htcentrum.pl

Abstract

Background:

It is unknown if continuous exposure to artificial intelligence (AI) changes endoscopists' behavior in colonoscopy. We assessed how endoscopists, who regularly used AI, performed colonoscopy once AI was unavailable.

Methods

We conducted a retrospective observational study at four endoscopy centers in Poland. These centers implemented AI tools for polyp detection at the end of 2021, after which colonoscopies were randomly conducted with or without AI assistance according to the date of examination. We evaluated the colonoscopy quality between September of 2021 and March of 2022, by comparing two different phases: three months before and after AI implementation. The primary outcome was adenoma detection rate (ADR) of standard, non-AI assisted colonoscopy. Multivariable logistic regression was done to identify independent factors affecting ADR.

Findings

A total of 1,443 patients underwent standard colonoscopy before (n=795) and after (n=648) AI implementation (mean age 58, 59% female). ADR of standard colonoscopy significantly decreased from 28.4% (226/795) before to 22.4% (145/648) after the exposure to AI, corresponding to a 6% absolute reduction (95% confidence interval: -10.5% to -1.6%). In multivariable logistic regression analysis, exposure to AI (0.69 [0.53 to 0.89]), patient's female sex (1.78 [1.38 to 2.30]), age < 60 (3.60 [2.74 to 4.72]), and alarm symptoms (1.36 [0.89 to 2.08]) were the independent factors affecting ADR.

Interpretation

Continuous exposure to AI may reduce ADR of standard colonoscopy, suggesting a negative impact on endoscopist behavior.

Funding

European Commission, Japan Society for the Promotion of Science, and Italian Association for Cancer Research.

Research in Context Panel

Evidence before this study

Many randomized trials show that the use of artificial intelligence (AI) for polyp detection can increase adenoma detection rate (ADR), a representative quality indicator of colonoscopy which is associated with colorectal cancer prevention. However, these is scarcity in research on how continuous exposure to AI affects endoscopist behavior including deskilling risk, although deskilling is of concern when implementing automation devices in general. We conducted a PubMed search from database inception to November 30, 2024, using the search terms "((endoscopists) AND (deskilling)) AND (artificial intelligence)", without applying language restrictions. While several guidelines and review articles cautioned the risk of deskilling, there was no relevant original research that investigated the deskilling risk associated with the use of AI in colonoscopy.

Added value of this study

We found ADR of standard, non-AI assisted, colonoscopy significantly decreased from 28.4% (226/795) before to 22.4% (145/648) after the exposure to AI, corresponding to a 6% absolute reduction or "deskilling". To our knowledge, this is the first study that suggests the negative impact of AI exposure on patient-relevant endpoints in medicine in general.

Implications of all the available evidence

The present study, despite an observational design being sensitive to selection bias and confounding, triggers the need for further high-quality studies in the AI area in gastroenterology and beyond given the adoption of AI in medicine is rapidly spreading. This emerging area of research is critical, especially in light of the medical principle "do no harm," which can be extended beyond patients to include the capability of physicians. Future studies may delve deeper into physician behavior, examining how AI affects clinical performance and identifying solutions to mitigate deskilling risk, including both computational solutions, such as explainable AI, and psychological interventions, such as cognitive forcing techniques.

Introduction

Colorectal cancer is a major healthcare problem.[1] Colonoscopy enables detection and removal of precancerous lesions (i.e. adenomas), leading to prevention of colorectal cancer.[2] Adenoma detection rate (ADR) - the proportion of colonoscopies in which one or more adenomas are detected – is a widely accepted indicator of a colonoscopist's performance with a higher ADR associated with a greater cancer prevention effect.[3] However, around one fourth of endoscopists do not achieve the recommended level of ADR according to a large-scale randomized trial.[4] Thus, maintaining a high ADR level is considered an important goal for both endoscopists, healthcare systems, and researchers.

Recently introduced computer-aided polyp detection powered by artificial intelligence (AI) may entail increasing ADR regardless of the expertise of endoscopists; a meta-analysis of 20 randomized trials showed an absolute 8.1 % increase in ADR with the use of AI during colonoscopy.[5] This AI-driven ADR increase is expected to improve colorectal cancer prevention effect. [6]

Despite its promise, it is unknown if continuous exposure to AI impacts endoscopists' performance during standard, non-AI assisted colonoscopy. This question is crucial as AI adoption in medicine is rapidly spreading. Psychological studies suggest that ongoing AI exposure may impact behavior in different ways: positively, by training clinicians, or negatively, through a "de-skilling" effect, where automation use leads to cognitive skill decay—patterns observed in non-medical fields.[7] To this end, we investigated changes in the quality of standard, non-AI assisted colonoscopy before and after the exposure to AI in routine practice.

Methods

Study design

We conducted a multicenter, retrospective, observational study at four Polish endoscopy centers. These centers implemented AI tools for polyp detection around the end of 2021 (25^{th} November of 2021 at Center 1, 14^{th} December of 2021 at Center 2, 7^{th} December of 2021 at Center 3, and 26^{th} November of 2021 at Center 4), after which colonoscopies were randomly done either with or without AI assistance according to the date of colonoscopy. We evaluated changes in the quality of all diagnostic, non-AI assisted colonoscopies between the 8^{th} of September of 2021 and 9^{th} of March of 2022, by comparing two different phases: approximately 3 months before and after AI implementation in clinical practice. The first 100 colonoscopies per center after AI implementation were excluded from analysis to allow endoscopists to become familiar with using AI. The institutional review board granted an exemption from review (No. ŚIL.KB.1159.2022) about the present study as it was observational and did not use any person-identifiable information. The study was conducted in accordance with the Declaration of Helsinki.

This study is nested in the ACCEPT project that investigates the effectiveness of AI tools in colorectal cancer screening programs.[8] ACCEPT is a randomized trial in which individuals undergoing screening colonoscopy are randomly assigned to either AI-assisted or standard colonoscopy, depending on the date of randomization. On AI days, the AI processor is automatically activated, while on non-AI days, it is automatically deactivated.

In Poland, participant consent for the ACCEPT study is waived due to the use of a routinely approved, minimally invasive medical device, the study's focus on quality improvement in cancer screening, and its nature as a health services implementation study (ethics approval number 57/2021).[9] Since the AI processor was automatically switched on or off, clinical patients also received either diagnostic colonoscopy with or without AI assistance, accordingly. Offering these health services to clinical patients was deemed ethically feasible, as both colonoscopy with and without AI assistance are part of routine clinical practice.

Participants

We collected relevant quality indicator data per colonoscopy including endoscopist's expertise and specialties (either physicians or surgeons), patients' age and sex, indication of colonoscopy, use of sedation, bowel preparation status (defined as adequate when Boston Bowel Preparation Score was six or more)[10], completeness of colonoscopy, and clinicopathological characteristics of the detected polyps. We excluded patients due to the following reasons: inability to undergo biopsy/polypectomy because of medication or coagulation disorders; pregnancy; referral for the assessment/treatment of known lesions; history of bowel resection; inflammatory bowel disease; colonoscopies performed with non-high definition colonoscopes. Endoscopists who did not conduct colonoscopy in one of the two study phases were also excluded. Data collection was done by KB and PK between April and October 2022.

Procedures

Colonoscopies were done with the high-definition endoscopy processors (EVIS X1 CV 1500, Olympus Medical Systems Corp., Tokyo) with compatible endoscopes (CF-H190L, CF-HQ190I, CF-HQ1100DI, PCF-H190I; Olympus Medical Systems Corp., Tokyo). The implemented AI system was ENDO-AID CADe (OIP-1, Olympus Medical Systems Corp., Tokyo) which helped detect polyps during colonoscopy. All the colonoscopies were performed as outpatient appointments. Patients were prepared with sodium picosulfate and magnesium citrate, low dose polyethylene glycol, or high dose polyethylene glycol. Preparation was assessed using Boston Bowel Preparation Scale.[10] Sedation was only administered on patient request, with no additional clinical or procedural criteria influencing the decision. Blinding the use of AI during colonoscopy was impossible due to the system's design, as the endoscopist could see the marked target on the screen during the AI use.

Outcomes

We primarily measured change in ADR of standard, non-AI assisted colonoscopy before and after the endoscopists were exposed to AI in clinical practice (primary outcome measure). In addition, we measured the change in the mean number of adenomas per colonoscopy and the mean number of advanced adenomas per colonoscopy before and after AI implementation.

Statistical analysis

All the data were analyzed by Stata version 18 (StataCorp LLC, Texas, US). Patient characteristics were described using mean and standard deviation (SE) for continuous and normally distributed variables, median and inter-quartile range (IQR) for skewed. Categorical variables were described using frequency and percentage. A p-value <0.05 was considered as statistically significant.

ADR was calculated as a proportion of colonoscopy procedures where one or more adenomas or cancers were detected. Sessile serrated lesions were not treated as adenomas while traditional serrated adenomas were treated as adenomas in the calculation of ADR. We did not take sessile serrated lesions (SSL) into ADR calculation to replicate the value of ADR in the previous large-scale observational studies which established ADR's association with interval colorectal cancers. [11,12] We included all the patients for ADR calculation including those with incomplete colonoscopy or inadequate bowel preparation. ADR of standard, non-AI assisted colonoscopy was compared between before and after the exposure to AI using a chi-squared test with % difference and 95% confidence interval provided. The mean number of adenomas and advanced adenomas per colonoscopy were compared between groups using a t-test, with mean difference, 95% CI and p-value provided.

Univariable mixed effects logistic regression [13,14] with random effect for endoscopists was carried out for detection of at least one adenoma as the outcome variable and the following potential predicting variables: patients' age and sex, use of sedation, bowel preparation status, completeness of colonoscopy, endoscopist specialty, endoscopist experience (years after graduation from medical school), endoscopist sex, center and implementation of AI in clinical practice. Variables with a p-value <0.05 in the univariate model were included in the multivariable model.

To account for potential imbalances in demographic characteristics, we did stratified analyses by sex and age of the patients and explored heterogenicity (female and male who were younger than 60 years old; female and male who were 60 years or older). Regarding age stratification, while recent American guidelines have lowered the screening age to 45 [15], we used 60 years as the cutoff in the subgroup analysis, as many studies and risk models favor this threshold due to a marked increase in adenoma prevalence.[16]

In addition, we compared ADR in the two phases (before and after AI implementation) by subgroups of center, specialties (physicians and surgeons) and endoscopist sex. In each subgroup, the percentage of ADR is given along with the absolute percentage change and 95% confidence interval (CI) comparing ADR before and after AI implementation. ADR of each endoscopist before and after AI exposure is also presented. As a supplementary analysis, univariable and multivariable mixed effects regression analysis including patients who underwent AI assisted colonoscopy during study period were performed.

Role of funding source

Funding source did not play any role in study design, data collection, data analysis, data interpretation or writing of the manuscript.

Results

In the observed period between the 8th of September in 2021 and 9th of March in 2022, a total of 2,177 colonoscopies were conducted, including 1,443 without AI use and 734 with AI. Our primary analysis focuses on the 1,443 patients who underwent standard, non-AI assisted colonoscopies before (795) and after (648) AI implementation. The colonoscopies were performed by 19 endoscopists who completed the endoscopic training (16 physicians and 3 general surgeons) with experience of over 2,000 colonoscopies each.

Patients who underwent standard, non-AI assisted colonoscopy after AI implementation were slightly younger (Median = 62 IQR [46, 70] before AI implementation vs median 59 IQR [44, 70] after AI implementation) with less females 54.6% compared to 62% before AI implementation, and a higher percentage of sedation usage (81.9% vs 77.1% before AI implementation) – see Table 1. Adequate bowel preparation rates were similar before and after AI implementation (96.6% vs 96.8%, p=0.870).

A total of 177 (12.3%) patients were referred to due to alarming symptoms (weight loss, anemia, gastrointestinal bleeding, tumor seen in computed tomography), 230 (15.9%) were surveillance colonoscopies, 32 (2.3%) were positive for fecal occult blood test (FOBT), 467 (32.4%) had other indications (change in bowel movements habit, diarrhea), and 537 (37.2%) were with unknown indications. Broadly similar percentages were seen between before and after AI implementation. The number of colonoscopies in which the cecum could not be reached was 4 (0.5%) before AI implementation and 8 (1.2%) after AI implementation.

ADR at standard, non-AI assisted colonoscopies significantly decreased from 28.4% (226/795) before AI exposure to 22.4% (145/648) after AI exposure, corresponding to a 6% absolute reduction (95%CI -10.5% to -1.6%, p=0.009, see Table 2 and Figure 1).

Mean adenoma per colonoscopy (APC) before AI exposure was slightly higher at a mean of 0.54 compared to 0.43 after AI exposure, yielding a mean difference of 0.11 (95% CI; -0.01 to 0.24) with p=0.071. Considering the number of APC in patients with at least one adenoma detected, this rate did not change significantly between the groups before and after AI exposure with mean 1.91 (SE 0.11) before versus mean 1.92 (SE 0.14) after AI exposure, resulting in a difference of -0.01 (95% CI -0.36 to 0.34, p=0.954.) Mean advanced adenoma per colonoscopy (AAPC) was similar between the two periods (mean 0.062 vs 0.063 mean difference -0.002 [-0.03, 0.03]; p=0.918). Mean AAPC on standard colonoscopy in patients with at least one adenoma detected was 0.22 (SE 0.03) before AI exposure and 0.28 (SE 0.05) after AI exposure with difference -0.06 (-0.18, 0.05, p=0.261). Colorectal cancers were detected in 6 colonoscopies (0.75%) before AI exposure and in 8 colonoscopies (1.2%) after AI exposure (p=0.35).

Univariable mixed effects logistic regression was performed to identify potential predictors of ADR, including exposure to AI, female sex, age<60, alarm symptoms. In multivariable logistic regression analysis, exposure to AI (0.69 [0.53 to 0.89]), patient’s female sex (1.78 [1.38 to 2.30]), age < 60 (3.60 [2.74 to 4.72]), and alarm symptoms (1.36 [0.89 to 2.08]) were the independent factors that affected ADR (see Table 3).

Endoscopist-level analysis showed that most endoscopists reduced their ADR when performing standard colonoscopies after the AI exposure, whereas only four endoscopists increased their ADR (see Figure 2 and Appendix p 2).

Among physicians ADR was reduced by 6.1% (95% CI -11.0%, -1.2%) from 29.7% to 23.5%, and among surgeon ADR was reduced by 8.3% (95% CI -18.9%, -2.3%) from 21.8% to 13.5% (see Appendix p 1).

When focusing on sex of the endoscopists, male endoscopists reduced their ADR by 2.9% (95% CI, -8.0% to 2.3%) from 25.2% (148/587) to 22.4% (106/474), while female endoscopists reduced their ADR by 15.1% (95%CI, -24.1% to -6.0%) from 37.5% (78/208) to 22.4% (39/174).

In all centers ADR for standard, non-AI assisted colonoscopy was reduced after the AI exposure, the magnitude of ADR reduction varied greatly between the centers (see Appendix p 3). Center 1 and 3 which had higher baseline ADR levels before the AI exposure (29.4% and 39.7%) showed a greater reduction in ADR (Centre 1: -8.4% (95% CI -17.4%, 0.6%), Centre 3: -12.1% (95% CI -21.4%, 2.8%)). Center 2 and 4 with moderate baseline ADR levels before the AI exposure (21.7% and 22.9%) showed a much smaller reduction in ADR (see Appendix p 3).

ADR of standard, non-AI assisted colonoscopy decreased after the AI exposure in women <60 years by -6.7% (95% CI -12.8%, -0.6%) from 14.0% to 7.3%, and in women ≥60 years by -8.1% (95% CI -16.5%, 0.3%) from 35.1% to 27.0%. Men < 60 years had a reduction of -8.0% with 95%CI (95% CI -16.9%, 0.8%), while those 60 and over years had a 1.6% increase in ADR after the AI exposure (95% CI -9.8%, 3.0%) (see Appendix p 4).

A comparison was also done on ADR between before and after AI exposure across different indications. ADR numerically decreased after AI exposure in all the five different indication groups with differences range between -4.3% and -20.8%. (see Appendix p 5)

ADR of AI-assisted colonoscopy was 25.3% (186/734). The multivariate logistic regression analysis showed that AI exposure, female sex and patient age < 60 were independent factors that affected ADR, while adjusted for endoscopist as a random effect. Compared to colonoscopies before AI implementation, use of AI did not significantly affect the ADR (see Appendix p 6).

Discussion

Many randomized trials show that AI use for polyp detection increases ADR by absolute 5-20%, generating enthusiasm for this technology among physicians, industry, and society [17]. However, this excitement has diverted attention from another key clinical question: the impact of AI on human capability. It remains unclear how continuous exposure to AI affects endoscopists' behavior and patient-relevant outcomes in standard, non-AI assisted procedures. This is important because psychological studies show that dependance on AI can influence decision-making and human behavior that may apply to colonoscopy practice. Previous research outside colonoscopy has shown that the use of AI can lead to over-reliance on technology,[18,19] a loss of domain expertise and a de-skilled work force in non-clinical tasks.[7,20] Our primary analysis showed continuous exposure to AI reduced ADR of standard, non-AI assisted colonoscopy from 28.4% to 22.4% with a 6% absolute difference, suggesting a detrimental effect on endoscopist capability. Notably, during AI implementation phase, colonoscopies involved more male patients (38.0% vs. 45.4%, P=0.005) and greater sedation use (77.1% vs. 81.9%, P=0.024)—both factors that would typically increase ADR. Yet, ADR declined after AI implementation, emphasizing the significance of this observed reduction.

Interpretation of this data is challenging. First, our observational study design necessitates cautious assessment because it is vulnerable to selection bias and confounding. To minimize the impact of confounders, we conducted stratified and multivariate analyses including risk factors of adenomas, all of which suggested that continuous exposure to AI reduces ADR. However, robustly designed prospective trials are warranted to generalize these findings.

Another challenge is to understand why the detrimental effect occurred. We assume that continuous exposure to decision support systems like AI may lead to the natural human tendency to over-rely on their recommendations, leading to clinicians becoming less motivated, less focused, and less responsible when making cognitive decisions without AI assistance.[21] In fact, the European Society of Gastrointestinal Endoscopy published caution about such risk of deskilling in their guideline recommendations in 2019.[22] A recently published experimental study also supports this interpretation by showing reduced visual eye-movements during colonoscopy when using AI for polyp detection, indicating a risk of overdependence.[23] In a similar study carried out in AI for breast cancer detection for mammography, physicians' detection capability significantly decreased when AI support was expected.[24] However, these early-stage experimental studies may require real-world clinical studies to validate findings using patient-important outcomes. To our knowledge, the present study is the first to examine the impact of continuous AI exposure on such outcomes, like ADR, in medicine.

The present study also gives a unique insight into previously published randomized trials in the field. [17,25] In these trials, AI-assisted colonoscopy provided a higher ADR than non-AI assisted colonoscopy which was considered the standard of care. However, we may doubt if it was truly the standard of care. Our study suggests that non-AI assisted colonoscopy assessed in these randomized trials could be different from everyday colonoscopy, because ADR of the non-AI assisted colonoscopy could be negatively affected by continuous AI exposure. This artificially reduced ADR level of non-AI assisted colonoscopies might help create significant differences of ADR in these randomized trials. Our results support this hypothesis. While AI-assisted colonoscopy appeared to increase ADR from 22.4% to 25.3% within the same time frame (see Appendix p 6), the actual trend from a chronological perspective suggests a decline in ADR for non-AI-assisted colonoscopy, dropping from 28.4% to 22.4%. This indicates that the increased

ADR shown in many randomized trials may, at least in part, be due to a reduction in ADR among non-AI procedures.

Our sub-group analyses also bring interesting findings. Centers with relatively modest baseline ADR (21.7% to 22.9%) showed a less pronounced deskilling effect compared to the centers with higher baseline ADR (29.4% to 39.7%), suggesting that lower baseline colonoscopy quality leaves less room for further ADR reduction. A large drop in ADR in the centres with the high ADRs before the AI implementation could be the result of regression to the mean. Furthermore, another sub-group analyses suggest that the detrimental effect may be more pronounced among female patients and by female endoscopists. Further research is needed to clarify these causal relationships.

The present study has several limitations. First, the compared patient cohorts might not have comparable background characteristics including indications and procedural contexts due to the observational study design. To mitigate this, we conducted multivariable logistic regression analysis. Second, only one AI system was used in this study, which may limit generalization of the study results. However, a randomized trial showed this device increased ADR by roughly relative 20% which is in line with the general performance of the similar AI devices.[17,26] Given the general nature of the AI tools and the tendency for humans to over-rely on them, we do not think the study results can apply only to this specific AI. Third, the findings are based on 19 "experienced" endoscopists participated in the study, which may limit its generalizability. Further studies involving less experienced operators are needed, as incorporating AI for training purposes is a major focus of the medical community.[27] Fourth, the number of procedures per endoscopist was insufficient to reliably assess endoscopist-level ADR changes. Thus, we treated endoscopist-level analysis as a subgroup analysis, advising cautious interpretation. Fifth, time effect could also influence endoscopists' skills, potentially hindering the interpretation of the study results. To minimize this bias, we focused on a limited timeframe (approximately 3 months before and after AI implementation) rather than year-level changes. Additionally, all participants were experienced endoscopists (average 27.6 years of experience, ranging from 8 to 39 years), making it unlikely that their skills would change significantly over a short period—thereby minimizing the risk of a time effect. Sixth, we primarily investigated the impact of AI exposure on human behavior; therefore, blinding was not done. However, the absence of blinding may introduce unforeseen biases and potentially distort the results. Seventh, the lack of data of endoscope withdrawal time and sample size calculation due to the retrospective observational design are also noticeable limitations. Eighth, we observed an almost two-fold increase in the number of procedures in the post-AI phase compared to the pre-AI phase (795 before vs. 1382 after AI implementation including 734 with AI-assistance and 648 without AI-assistance), even though both phases covered approximately three months. This was primarily due to the temporary suspension of the colonoscopy-based colorectal cancer screening programme in Poland on 1 January 2022 (lasting until December 2022). [28] This suspension has increased each centre's capacity for diagnostic colonoscopies, enabling them to accept more patients from the waiting list in the second phase, being potentially a cofounding factor. Ninth, exclusion of SSL from the ADR calculation may not accurately reflect colonoscopy quality as a recent Dutch study showed SSL detection rate was associated with post-colonoscopy colorectal cancer incidence.[29] Tenth, the first 100 colonoscopies per center following AI implementation were excluded to give endoscopists time to become familiar with the technology, although the number of cases performed by each endoscopist during this period may have varied. Finally, the number of examinations performed by some endoscopists was relatively low (see Appendix p 2), due to the observational nature of the study and could potentially affect the results.

In conclusion, we observed that continuous exposure to AI in colonoscopy may reduce the ADR of standard, non-AI assisted colonoscopy. We emphasize the urgent need for robust prospective studies such as randomized crossover trials to assess its generalizability and call for more behavioral research to understand the currently under-investigated mechanisms of how AI affects physician capability.

Funding

YM: European Commission (Horizon Europe: 101057099); Japan Society for the Promotion of Science (Number 22H03357),

CH: is supported by the European Commission (Horizon Europe 101057099). The Associazione Italiana per la Ricerca sul Cancro (AIRC): IG 2022 – ID. 27843 project / (AIRC) IG 2023 – ID. 29220 project and Bando PNRR-MCNT2-2023-12377041.

Declaration of interests

YM: Olympus Corp (consultation, speaking fees, device loan) and Cybernet System (loyalty fee), also supported by the European Commission (Horizon Europe: 101057099); Japan Society for the Promotion of Science (Number 22H03357),

MP: Received consultancy fee from Olympus Europe, Alfasigma and speaker's fee from Olympus, Casen Recordati, Mayoli and travels fee from Mayoli. Member of Chair of Equity and Diversity WG -ESGE, Spanish Gastroenterology Association President

CH: Odin (consultation, device loan), Olympus (device loan and speaking fees), also supported by the European Commission (Horizon Europe 101057099). The Associazione Italiana per la Ricerca sul Cancro (AIRC): IG 2022 – ID. 27843 project / (AIRC) IG 2023 – ID. 29220 project and Bando PNRR-MCNT2-2023-12377041.

SF: SF institution received funds to cover statistical consultancy activities.

M.F.K: Received consultancy fee from: Olympus, Boston Scientific, ERBE also Speaking and teaching fee from Olympus Corp, Fujifilm, Boston Scientific, Medtronic, ERBE, Microtech, Norgine.

K.B., M.R., D.K., P.K., M.B., H.O.A., J.B., M.Bu., N.H., T.R., Ø.H., K.J., M.Ku., N.P., M.Ka, M.Br. - declares no competing interests.

Author Contributions

Conceptualization: K.B., M.R., P.K., M.B., Y.M.

Data Curation: K.B., M.R., D.K., P.K., M.B., M.Bu., T.R., K.J., Y.M.

Formal Analysis: K.B., M.R., D.K., P.K., M.B., J.B., N.H., S.F., Ø.H., Y.M.

Investigation: K.B., M.R., P.K., M.B., M.Bu., T.R., K.J., Y.M.

Methodology: K.B., M.R., P.K., M.B., C.H., M.Ku., M.P., N.P., M.F.K., M.Ka., M.Br., Y.M.

Project Administration: K.B., M.R., P.K., M.B., C.H., M.Ku., M.P., N.P., M.F.K., M.Ka., M.Br., Y.M.

Resources: K.B., M.R., P.K., M.B., M.Bu., T.R., K.J., C.H., S.F., M.Ku., M.P., N.P., M.F.K., M.Ka., M.Br., Y.M.

Supervision: M.R., H.O.A., J.B., N.H., Ø.H., C.H., S.F., M.Ku., M.P., N.P., M.F.K., M.Ka., M.Br., Y.M.

Validation: M.R., H.O.A., J.B., N.H., Ø.H., C.H., S.F., M.Ku., M.P., N.P., M.F.K., M.Ka., M.Br., Y.M.

Visualization: M.R., Y.M.

Writing – Original Draft: K.B., M.R., D.K., Y.M.

Writing – Review & Editing: K.B., M.R., H.O.A., Ø.H., S.F., C.H., M.Ku., M.P., N.P., M.F.K., M.Ka., M.Br., Y.M.

Data sharing statement

Data and analytic codes can be shared under the following conditions: 1. Approval of all the data providers/owner. 2. Signed agreement for data transfer. 3. Approval of relevant ethics committees. The corresponding author should be contacted with regard to the data sharing.

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Table 1. Characteristics of the patients who underwent standard, non-AI assisted colonoscopies

	Before AI implementation	**After AI implementation**	**Total**	**p-value**
N (%)	795 (55.1)	648 (44.9)	1443	
Age in years: median (IQR)	62 (46, 70)	59 (44, 70)	61 (45, 70)	0.070
Male: n (%)	302 (38.0)	294 (45.4)	596 (41.3)	0.005
Sedation: n (%)	613 (77.1)	531 (81.9)	1144 (79.3)	0.024
Adequate bowel preparation*: n (%)	768 (96.6)	627 (96.8)	1395 (96.7)	0.870
Incomplete examination	4 (0.5)	8 (1.2)	12 (0.8)	n/a*****
Indications n (%)				0.009
Alarm symptoms**	99 (12.5)	78 (12.0)	177 (12.3)	
Surveillance	119 (15.0)	111 (17.1)	230 (15.9)	
FOBT+***	21 (2.6)	11 (1.7)	32 (2.2)	
Other****	277 (34.8)	190 (29.3)	467 (32.4)	
Unknown	279 (35.1)	258 (39.8)	537 (37.2)	

AI-Artificial intelligence; IQR- Interquartile range; *Adequate preparation understood by score of at least 6 in Boston Bowel Preparation Score; [10] ** Weight loss, anemia, gastrointestinal bleeding signs and tumor seen in computed tomography scan; *** Positive fecal occult blood test **** Change in bowel movement habits, diarrhea. ***** Significance assessment was not done due to too small number of events.

Table 2. Quality indicators of standard, non-AI assisted colonoscopy before and after the artificial intelligence implementation.

	Before AI implementation N=795	After AI implementation N=648	Difference (95% CI); p-value
ADR n (%)	226 (28.4%)	145 (22.4%)	-6.0 (-10.5, -1.6); p = 0.009#
Adenoma per colonoscopy, mean (SE)	0.54 (0.04)	0.43 (0.04)	0.11 (-0.01, 0.24); p = 0.071~
Advanced adenomas per colonoscopy, mean (SE)	0.062 (0.01)	0.063 (0.01)	-0.002 (-0.03, 0.03); p=0.918~

AI-artificial intelligence; ADR-Adenoma detection rate; SE standard error; #chi-squared test; ~t-test

Table 3. Logistic regression analysis to identify factors affecting adenoma detection rate.

Variable		Adenoma n/N (%)	Univariable Odds ratio (95% CI), p-value	Multivariable Odds ratio (95% CI), p-value#
AI implementation in clinical practice (exposure to AI)	Before	226/795 (28.4%)	-	-
	After	145/648 (22.4%)	0.70(0.55, 0.90), p = 0.005	0.69 (0.53, 0.89), p = 0.004
Sex	Female	191/847 (22.6%)	-	-
	Male	180/596 (30.2%)	1.51 (1.19, 1.93), p = 0.001	1.78 (1.38, 2.30), p<0.001
Age group	<60 years	102/687 (14.9%)	-	-
	≥60 years	269/756 (35.6%)	3.46 (2.65, 4.51), p <0.001	3.60 (2.74, 4.72), p<0.001
Adequate bowel preparation Preparation*	No	9/48 (18.8%)	-	-
	Yes	362/1395 (26.0%)	1.49 (0.70, 3.14), p = 0.299	-
Incomplete examination	No	369/1431 (25.8%)	-	-
	Yes	2/12 (16.7%) **	-	-
Sedation	No	75/299 (25.1%)	-	-
	Yes	296/1144 (25.9%)	0.89 (0.63, 1.27), p = 0.531	-
Indication	Alarm symptoms	36/177 (20.3%)	-	-
	Surveillance/Other/ FOBT+*** Unknown	335/1266 (26.5%)	1.67 (1.12, 2.51); p = 0.013	1.36 (0.89, 2.08); p = 0.154
Specialty	Physician	334/1245 (26.8%)	-	-
	Surgeon	37/198 (18.7%)	0.61 (0.35, 1.08); p = 0.088	-
Centre	1	91/354 (25.7%)	-	-
	2	66/308 (21.4%)	0.78 (0.50, 1.23); p = 0.285	-
	3	132/391 (33.8%)	1.37 (0.88, 2.13); p = 0.154	-
	4	82/390 (21.0%)	0.77 (0.47, 1.1.26); p = 0.293	-
Endoscopist sex	Male	254/1061 (23.9%)	-	-
	Female	117/382 (30.6%)	1.14 (0.71, 1.85); p = 0.588	-
Endoscopist experience	Per year since graduation	-	0.98 (0.96, 1.01); p = 0.245	-

AI-artificial intelligence; CI confidence interval; *Adequate preparation understood by score of at least 6 in Boston Bowel Preparation Scale;[10] #Kept in model if univariable p<0.05. All models include endoscopists as a random effect[13,14].**Significance assessment was not done due to too small number of events. *** Positive fecal occult blood test

Appendix

Supplement to: Endoscopist de-skilling risk after exposure to artificial intelligence in colonoscopy: a multicenter observational study

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Supplementary Table 1. Comparison of adenoma detection rate of standard, non-AI assisted colonoscopy between different endoscopists' specialties.

Specialty (n)	Total colonoscopies	Before the AI implementation % (n/N)	After the AI implementation % (n/N)	% difference (95% CI)
Physicians (n=16)	1245	29.7% (199/671)	23.5% (135/574)	-6.1% (-11.0%, -1.2%)
Surgeons (n=3)	198	21.8% (27/124)	13.5% (10/74)	-8.3% (-18.9%, 2.4%)

AI- Artificial intelligence; CI confidence interval; Data presented is with a focus on 16 physicians and 3 surgeons included in the present study

Supplementary Table 2. Comparison of endoscopist's adenoma detection rate of standard, non-AI assisted colonoscopy before and after AI implementation.

Endoscopist	Sex	Centre	Specialty	Years of experience	Total colonoscopies	Before the AI implementation % (n/N)	After the AI implementation % (n/N)
1	F	3	Physician	15	66	56.7% (17/30)	19.4% (7/36)
2	M	3	Physician	27	72	19.1% (8/42)	23.3% (7/30)
3	M	1	Physician	8	48	33.3% (7/21)	25.9% (7/27)
4	M	4	Physician	31	160	27.6% (23/86)	18.9% (14/74)
5	M	3	Physician	39	12	0% (0/4)	25.0% (2/8)
6	M	1	Physician	21	115	29.5% (18/61)	20.4% (11/54)
7	M	2	Surgeon	31	79	36.2% (17/47)	15.6.% (5/32)
8	F	2	Physician	29	20	5.9% (1/17)	0.0% (0/3)
9	F	3	Physician	17	153	50.0% (41/82)	28.2% (20/71)
10	M	2	Physician	36	49	25% (8/32)	35.3% (6/17)
11	M	2	Physician	24	51	31.3% (10/32)	21.1% (4/19)
12	M	2	Physician	34	7	40% (2/5)	100% (2/2)
13	M	1	Physician	35	48	38.9% (14/36)	25% (3/12)
14	F	1	Surgeon	30	32	23.8% (5/21)	9.1% (1/11)
15	M	4	Physician	32	230	20% (23/115)	19.1% (22/115)
16	M	3	Physician	33	88	31.7% (13/41)	36.2% (17/47)
17	F	1	Physician	23	111	24.1% (14/58)	20.8% (11/53)
18	M	2	Physician	32	15	0% (0/9)	33.3% (2/6)
19	M	2	Surgeon	27	87	8.9% (5/56)	12.9% (4/31)

AI Artificial intelligence: n = number of colonoscopies with adenoma detected, N = total number of colonoscopies

Supplementary Table 3. Comparison of adenoma detection rate of standard, non-AI assisted colonoscopy between the study centers.

Centre (n)	Total colonoscopies	Before the AI implementation % (n/N)	After the AI implementation % (n/N)	% difference (95% CI)
1	354	29.4% (58/197)	21.0% (33/157)	-8.4% (-17.4%, 0.6%)
2	308	21.7% (43/198)	20.9% (23/110)	-0.8% (-10.3%, 8.7%)
3	391	39.7% (79/199)	27.6% (53/192)	-12.1% (-21.4%, 2.8%)
4	390	22.9% (46/201)	19.1% (36/189)	-3.8% (-11.9%, 4.2%)

AI- Artificial intelligence; CI confidence interval

Supplementary Table 4. Comparison of adenoma detection rate of standard, non-AI assisted colonoscopy between patient sex and age group

Subgroup (n)	Total colonoscopies	Before the AI implementation % (n/N)	After the AI implementation % (n/N)	% difference (95% CI)
Female ≥ 60	468	35.1% (98/279)	27.0% (51/189)	-8.1% (-16.5%, 0.3%)
Male ≥ 60	288	40.9% (63/154)	42.5% (57/134)	1.6% (-9.8%, 3.0%)
Female <60	379	14.0% (30/214)	7.3% (12/165)	-6.7% (-12.8%, -0.6%)
Male <60	308	23.6% (35/148)	15.6% (25/160)	-8.0% (-16.9%, 0.8%)

AI- Artificial intelligence; CI confidence interval

Supplementary Table 5. Comparison of adenoma detection rate of standard, non-AI assisted colonoscopy between different indications groups

Indication (n)	Total colonoscopies	Before the AI implementation % (n/N)	After the AI implementation % (n/N)	% difference (95% CI)
Alarm symptoms	177	22.2% (22/99)	17.9% (14/78)	-4.3% (-16.0%, 7.5%)
Surveillance	230	47.9% (57/119)	37.8% (42/111)	-10.1% (-22.8%, 2.7%)
FOBT +	32	57.1% (12/21)	36.4% (4/11)	-20.8% (-56.2%, 14.7%)
Other Indications	499	23.1% (64/277)	19.0% (36/190)	-4.2% (-11.6%,3.3%)
Lack of indication	537	25.4% (71/279)	19.0% (49/258)	-6.4% (-13.4%, 0.5%)

AI- Artificial intelligence; CI-confidence interval, FOBT-Fecal Occult Blood Test

Supplementary Table 6. Logistic regression analysis to identify factors affecting adenoma detection rate in all 2177 colonoscopies (including those which used AI)

Variable		Adenoma n/N (%)	Univariable Odds ratio (95% CI)	Multivariable Odds ratio (95% CI), p-value#
AI implementation in clinical practice	Before AI implementation (without using AI in colonoscopy)	226/795 (28.4%)	1.00	1.00
	After AI implementation (without using AI in colonoscopy)	145/648 (22.4%)	0.70 (0.54, 0.89)	0.68 (0.53. 0.88)
	After AI implementation (using AI in colonoscopy)	186/734 (25.3%)	0.84 (0.67, 1.06)	0.80 (0.63, 1.02)
Sex	Female	273/1258 (21.7%)	1.00	1.00
	Male	284/919 (30.9%)	1.66 (1.36, 2.02)	1.89 (1.53, 2.32)
Age group	<60 years	164/1034 (15.9%)	1.00	1.00
	>60 years	393/1143 (34.4%)	3.06 (2.47, 3.79)	3.23 (2.60, 4.02)
Adequate bowel preparation Preparation*	No	14/68 (20.6%)	1.00	-
	Yes	543/2109 (25.8%)	1.37 (0.75, 2.52)	-
Sedation	No	110/446 (24.7%)	1.00	-
	Yes	447/1731 (25.8%)	0.90 (0.67, 1.22)	-
Indication	Alarm symptoms	52/260 (20.0%)	1.00	-
	Surveillance/Other/Unknown	505/1917 (26.3%)	1.713 (1.1.23, 2.39)	-
Specialty	Physician	504/1899 (26.5%)	1.00	-
	Surgeon	53/278 (19.1%)	0.64 (0.37, 1.11)	-
Centre	1	136/522 (26.1%)	1.00	
	2	94/446 (21.1%)	0.73 (0.47, 1.14)	
	3	206/613 (33.6%)	1.34 (0.87, 2.06)	
	4	121/596 (20.3%)	0.73 (0.43, 1.21)	
Endoscopist sex	Male	384/1610 (23.9%)	1.00	-
	Female	173/567 (30.5%)	1.11 (0.68, 1.79)	-
Endoscopist experience	Years since graduation from medical school	-	0.98 (0.96, 1.01); p = 0.163	-

AI-artificial intelligence; CI confidence interval; *Adequate preparation understood by score of at least 6 in Boston Bowel Preparation Scale;[1] #Kept in model if univariable p<0.05. All models include endoscopists as a random effect [2,3].

Multi-Centre Study List.

Centre name*	Number of patients
1	354
2	308
3	391
4	390

*The centre names correspond to those listed in Supplementary Table 3. We present the specific data for each of the four hospitals in an anonymized manner. These hospitals, referred to as Centres 1–4, include: Endoterapia, H-T. Centrum Medyczne, Tychy, Poland (PI: Tomasz Romańczyk); Centre of Gastroenterology, Wodzisław Śląski, Poland (PI: Krzysztof Jarus); Endomed, Radom, Poland (PI: Paweł Kołodziej); and Buszkiewicz Medical Center, Gorzów Wielkopolski, Poland (PI: Marek Buszkiewicz).

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Figure 1. Change in adenoma detection rate of standard, non-AI assisted colonoscopy before and after implementation of artificial intelligence for polyp detection. Each plot represents a 95% confidence interval.

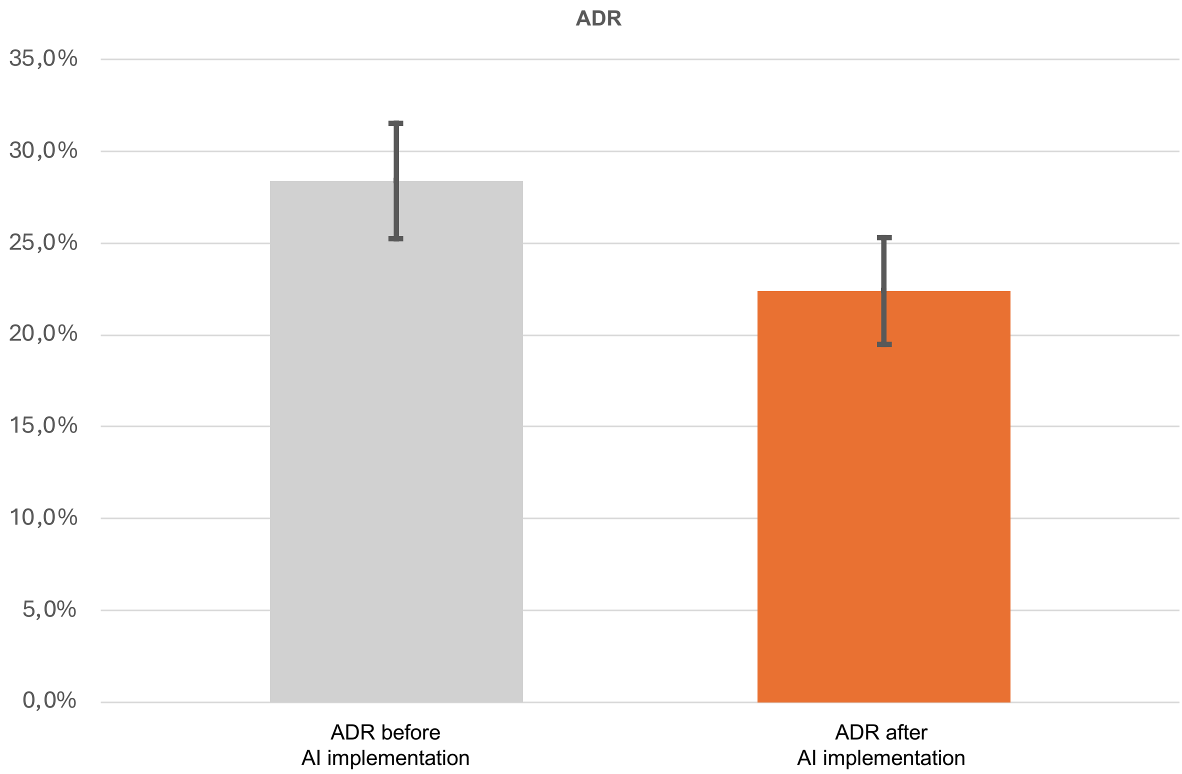

Figure 2: Endoscopist-level absolute percentage change of adenoma detection rate of standard, non-artificial intelligence (AI) assisted colonoscopy after implementation of AI in colonoscopy at each center. Four endoscopists who did less than 10 colonoscopies for either before or after AI implementation are removed from the presentation.